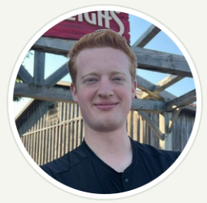
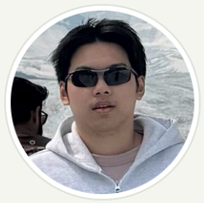
KINETIC · CSC454 / CSC491 · DEMO 1

# Emergency Transport Coordinator



Azaria Kelman



Richard Xu

# Every minute matters.



# For the patient, arriving is not the end of the wait

- 911 call
- Patient assessment
- Suitable hospitals
- Destination decision
- Transport
- ARRIVAL / DROP-OFF

**WAITING FOR CARE**

- Hospital care begins

THE BOTTLENECK

# At the hospital. Still waiting for care.

Hospital arrival ≠ care begins

# What if offload time became part of the destination decision?

**CURRENT**

Suitable hospitals

↓

Current information

↓

Choose destination

↓

Travel

↓

Unknown offload wait

**TARGET**

Suitable hospitals

↓

**Travel + hospital state**
**+ inbound ambulances + predicted offload**

↓

Compare destinations

↓

**Human chooses**

↓

Handover sooner → return to service

Compare clinically appropriate hospitals by travel + expected offload.

**The user makes the final decision.**

Demo | Try it live

HTML copy of the Overview · rankings from the app's simulation code · synthetic data

# Change the data source. Keep the workflow.

PROPOSED ARCHITECTURE

FRONTEND

**Next.js**

Patient overview and dispatcher views

REST · JSON

BACKEND · API

**FastAPI**

Used Python to help with modelling

SQL

DATA LAYER

**PostgreSQL**

Local database: runs, events, hospital state

### 🏥 Hospital side

- Eligibility rules
- Offload prediction
- Queue + inbound state

### 🚑 Ambulance side

- Hospital ranking
- Decision endpoint
- Fleet status

↑

**Simulator**

Synthetic calls and hospital reports today

**Operational feeds**

Future, through an adapter

The real app · Evaluation page · matched seeds, nearest vs coordinated

Connecting…

kinetic
AMBULANCE FLOW

K Team Kinetic
Research workspace

WORKSPACE

- Overview
- Scenario builder
- Live operations
- Hospital network
- Evaluation

About this workspace

TK Team Kinetic
CSC491 · Research & development

Workspace > Evaluation

SIMULATION MODE TK

# Evaluation

Synthetic data only

## Compare routing strategies

Nearest vs coordinated · identical calls per seed.

Scenario: Toronto · Morning shift

Distinct seeds: 5 seeds · standard

Run comparison

### Strategy comparison

Toronto · Morning shift · 5 seeds · 4 hours per run

Export results

**Mean offload**

Nearest 33.6 min
Coordinated 21 min

Δ -12.6 min
95% paired t interval: -17.6 to -7.7

**90th percentile offload**

Nearest 86 min
Coordinated 29.4 min

Δ -56.6 min
95% paired t interval: -83.1 to -30.1

**Mean travel**

Nearest 7.2 min
Coordinated 10.1 min

Δ +2.9 min
95% paired t interval: 2.2 to 3.6

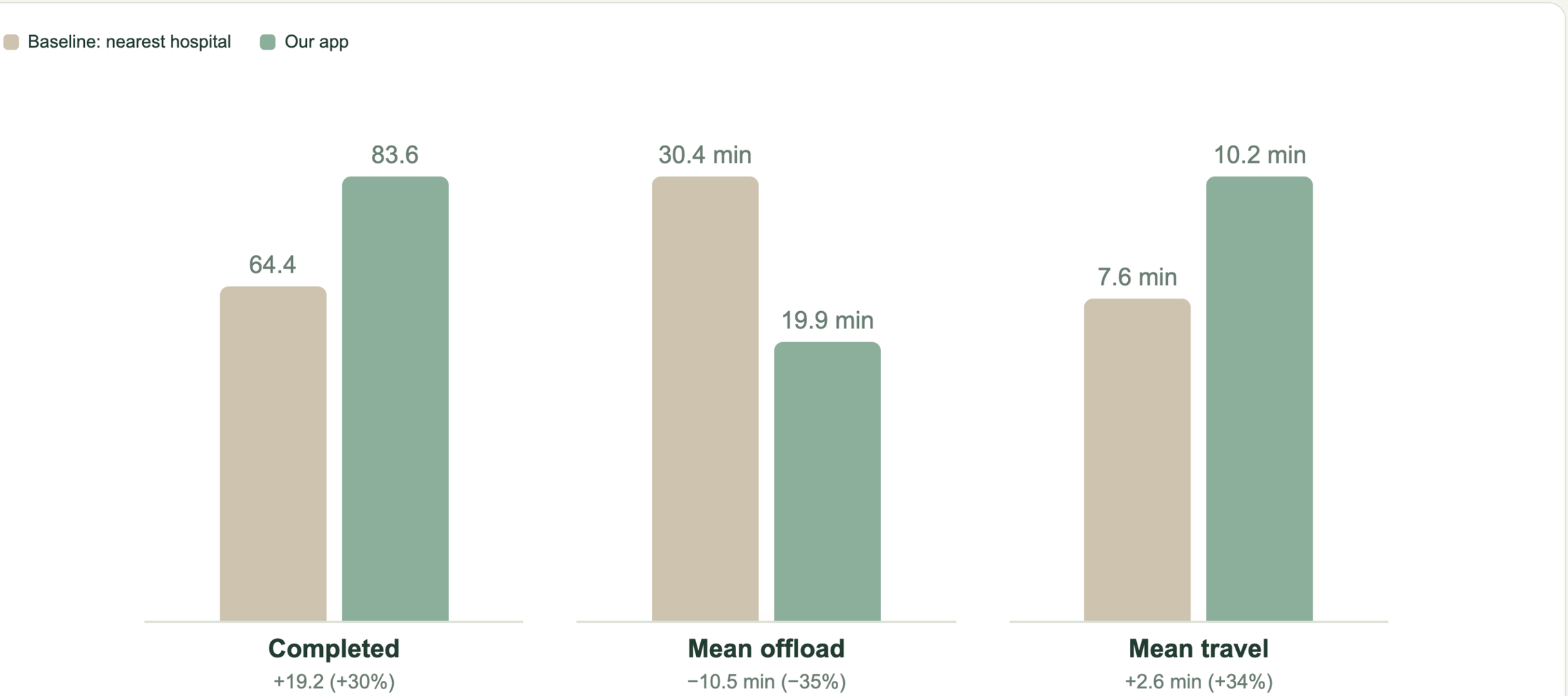
**Fleet availability**

Nearest 12.5 %
Coordinated 14.7 %

Δ +2.2 pp
95% paired t interval: -1.4 to 5.9

Synthetic results. Time averages exclude unfinished cases and may be biased.

# Early simulation results




We used a Python simulation with synthetic data (mean of 5 simulated shifts). Later on we will improve the data.

# The boundary of our proof

## Shown in Our MVP

✓ End-to-end data flow

✓ Dynamic state updates

✓ Destination comparison

✓ Quantitative evaluation

## Not proven

× Actual Ontario operational data

× Patient Distribution System / hospital integration

× Clinical outcome improvement

Next: real operational data + EMS partners.

KINETIC · EMERGENCY TRANSPORT COORDINATOR

# Thank you

Less time waiting at the hospital.

Questions?